

Control of a Multilevel Inverter based Grid Integration of Solar PV with BESS

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Abstract—In this paper, a multilevel inverter (MLI) based grid integration of solar photovoltaic (PV) with a battery energy storage system (BESS) is presented. The maximum power from the PV is extracted using an incremental conductance algorithm. The cascade proportional-integral (PI) control helps in the charging and discharging rate control of battery along with dc-link voltage regulation. An MLI capable of generating increased output voltage levels with the lower component count at reduced switching frequency increases efficiency. Power quality is enhanced at the point of interconnection of the system. Irrespective of the PV availability, a continuous, uninterrupted power supply is achieved with the help of BESS, making a reliable system. Hence, the proposed system is highly suitable for residential as well as commercial applications. This scheme feeds continuous power to the load along with the grid. The dynamic conditions occurring in the system apart from the steady-state conditions are analyzed using Matlab/ Simulink.

Index Terms—BESS, MPPT, Multilevel inverter, Power quality.

I. INTRODUCTION

Increased concern about the environment and the escalating power demand made the world to choose the path of power generation from renewable sources. There are different sources like solar, wind, geothermal, biomass, etc. However the major participants from all these are solar and wind [1]. Even though the potential is high, the primary issue to be tackled is the intermittent power generation capability.

Solar photovoltaic (PV) is one of the significant candidates, and here, the energy from the sun is used to produce power. Hence there are many factors that make the power production fluctuating. Environmental factors, solar insolation level, and temperature, are the key factors. So technology is necessary to extract the maximum power. Maximum power point tracking (MPPT) techniques are used for this purpose [2]. There are several MPPT algorithms reported in the literature. Perturb and observe (P&O), incremental conductance, fuzzy logic control etc are some of the major algorithm among them [3]- [5]. Even if there is a little compromise for accuracy, the reduced complexity, ease of implementation and robustness, P&O and incremental conductance are the top picked algorithms.

When there is a need to address critical loads and increase the system's reliability and stability, a storage system is necessary to overcome the intermittency issues caused by

renewable sources. Usually, battery energy storage systems (BESS) are used to serve this purpose. There are various configurations for BESS like single-stage system, two-stage system, etc. [6]. In a single-stage system, the battery is directly connected to the dc-link. However, here, the variations caused in the dc-link voltage directly affect the battery, which harms the performance and life of the battery. So to solve this problem two stage systems are used where bidirectional DC-DC converters (BDC) are used to interface the battery to the dc-link. Moreover, the BDC can be chosen such that the required criteria are satisfied. There are different configurations of BDC reported in state of the art like bidirectional buck-boost, Cuk, SEPIC, etc. [7]. A converter that can be made with fewer components with higher performances is reported in [8].

For the grid-connected applications, a DC-AC conversion stage is required to obtain an AC voltage. The increased penetration of power electronic converters and other nonlinear loads attenuates the grid power quality. So the inverter used for the DC-AC conversion should be capable of multi-functional capabilities so that the power quality is maintained. The conventional grid-connected two level inverters have many drawbacks like increased voltage stress, high dv/dt , increased total harmonic distortion (THD) in the output, the requirement of larger filter size, etc. Multilevel inverters (MLI) emerged as a better solution to tackle these drawbacks. There are different MLI topologies reported in the literature [9]. A reduced device count topology is the preferred one [10]. MLI can be operated as distribution static compensator (DSTATCOM), which operates on the distribution side and clears the power quality issues inflicted onto the grid side by the nonlinear load. DSTATCOM functionally works as a shunt active filter and the overall performance is enhanced.

The main objectives of this work are

- 1) Deploying BESS through a high gain bidirectional converter to ensure the bidirectional power flow.
- 2) Using MLI for the grid connection to overcome the challenges with conventional inverter.
- 3) Mitigation of harmonics and reactive power compensation at the point of interconnection.

This paper is sorted as follows; Section I deals with intro-

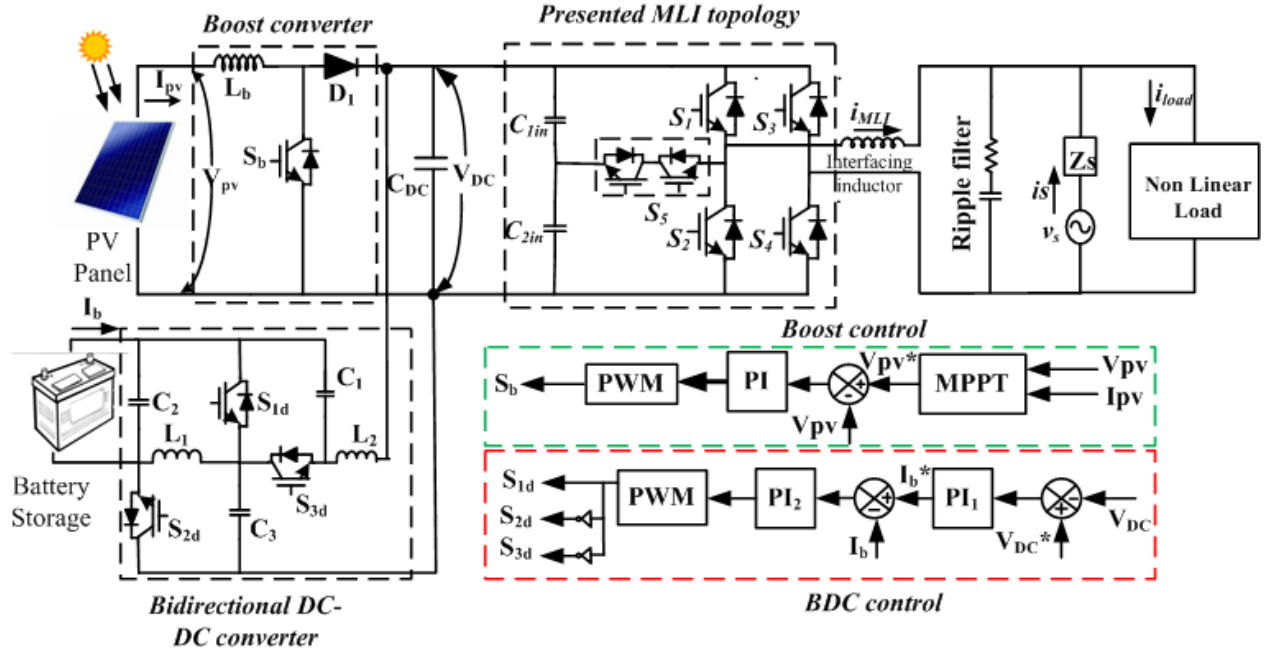


Fig. 1: Schematic diagram for system configuration

duction, Section II discusses system configuration followed by System control in section III. The simulation results are described in section IV and the conclusions in section V.

II. SYSTEM LAYOUT

The schematic layout for the proposed system is displayed in Fig. 1. The system involves solar PV as the renewable source. The MPPT is achieved through a boost DC-DC converter. An incremental conductance algorithm is used for peak power tracking. The operating voltage and current of PV array is 310 V and 14.6 A at standard test conditions. A battery bank is used as the energy storage stage. It is interfaced to the dc-link with the help of a bidirectional DC-DC converter [8]. The converter provides a higher voltage gain so that the battery voltage is considered as a lower value. This system is interfaced with the single-phase utility grid through a multilevel inverter, capable of producing five levels in the output with reduced device count. An RC ripple-filter is used to filter out the switching harmonics, and an interfacing inductor is connected to establish the power transfer and to eliminate the current ripples.

The dc-link voltage is maintained by BDC control and this function shifts with the inverter control in the absence of PV and battery. The charging and discharging of the battery depends on the available PV power and the load conditions. A fixed power is injected to the grid by the MLI so that it executes the function of active power injection. Moreover, the MLI is capable of meeting the load's reactive power requirement so that the grid can be maintained at unity power factor (UPF) operation.

A. Bidirectional DC-DC Converter

To ensure the bidirectional power flow between the storage unit and the dc-link, BDC is used. Here, a BDC capable of giving higher voltage gain with reduced components is employed so that battery with low voltage rating can be used [8]. The circuit configuration is shown in Fig. 1, where three switches are used. Voltage gain for the boost mode is given as

$$\frac{V_o}{V_{in}} = \frac{2 \times d}{1 - d} \quad (1)$$

Where d is the duty ratio of switch S_{1d} and the remaining two switches operate with the complementary pulse signal of switch S_{1d} .

B. Multilevel Inverter

The grid interfaced MLI is capable of producing five levels in the output with reduced number of switches when compared to other topologies in the literature [10]. Different modes of operation is shown in Fig. 2. Fig. 2(a) shows the conduction path for the positive voltage level $V/2$ where as Fig. 2(b) indicates the path for negative $V/2$. In both these conduction paths, the bidirectional switch S_5 is used. And the zero voltage

TABLE I: Switching states for different DC level

Level	S_1	S_2	S_3	S_4	S_5
V	1	0	0	1	0
$V/2$	0	0	0	1	1
0	1	0	1	0	0
$-V/2$	0	0	1	0	1
$-V$	0	1	1	0	0

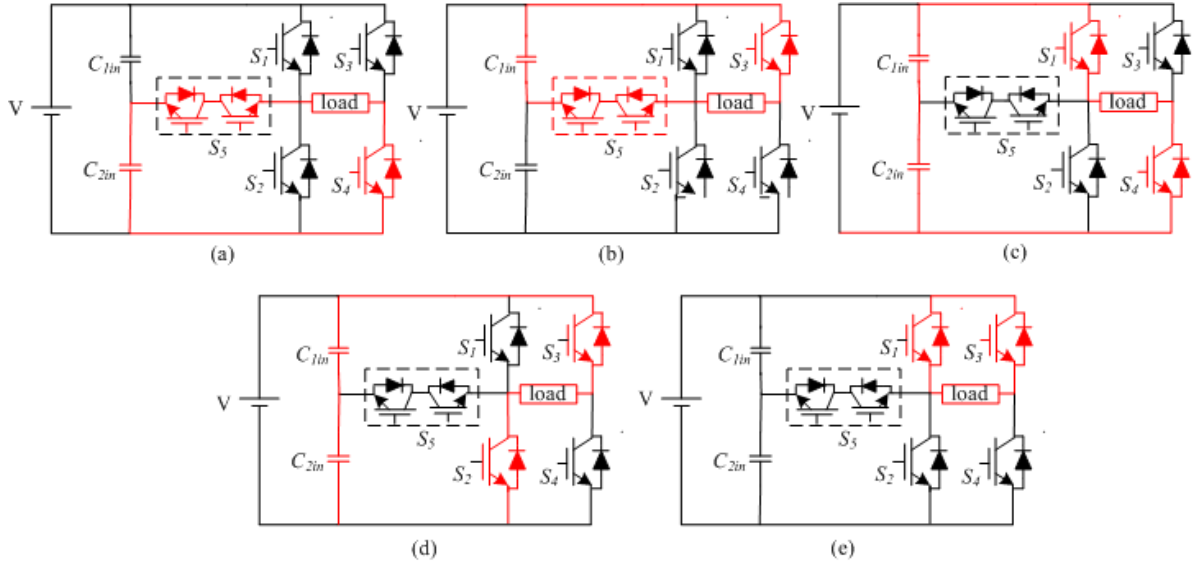


Fig. 2: Modes of operation of MLI (a) mode 1, (b) mode 2, (c) mode 3, (d) mode 4, (e) mode 5

level conduction path is shown in Fig. 2(e). The switching states for each DC level is organised in Table 1. The switching logic is selected so that the voltage balance occurs. So the self balancing of voltage across capacitors (C_{1in} and C_{2in}) are ensured without any external control circuit.

III. SYSTEM CONTROL

The control strategy for this system can be implemented into different sections which consist of MPPT control, BDC control and the MLI control.

A. MPPT with Voltage Control of Boost Converter

Incremental conductance algorithm is used to extract peak power from the PV panel [4]. Here, the PV voltage and current are sensed and the operating point corresponding to the peak power is obtained by taking the incremental conductance and the instantaneous conductance and the peak power is reached when they are equal. Fig. 3 shows the flowchart for the incremental conductance algorithm. The boost converter is controlled based on the reference PV voltage which is the voltage corresponding to the MPP and the actual value. Then the error signal is passed through a proportional integral (PI) controller which then passed to pulse width modulation (PWM) block to produce the switching pulse. The switching frequency for the boost converter is selected as 10 kHz.

$$e_b = (V_{PV}^* - V_{PV}) \times (k_{pb} + K_{ib}/s) \quad (2)$$

where k_{pb} and k_{ib} are the proportional (P) and integral (I) controller gains of the PI controller, and e_b is the output from the PI controller. The control diagram is shown in Fig. 1.

B. Bidirectional Converter Control

The switching pulses for the BDC is obtained after a cascade control consisting of dc-link regulation and battery current control. The control diagram for the BDC is shown in Fig.

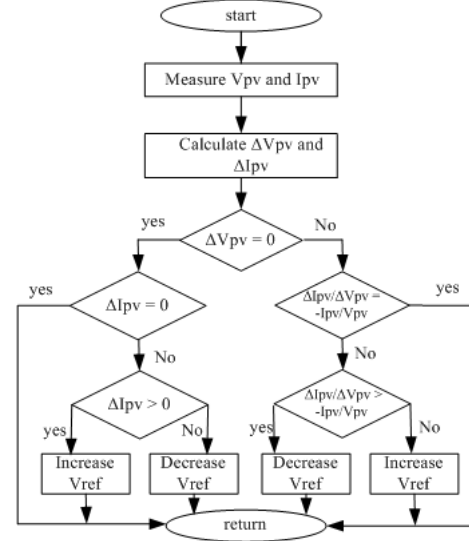


Fig. 3: Flowchart of incremental conductance algorithm

1. The dc-link voltage(V_{DC}) is controlled to reference dc-link value(V_{DC}^*), which is then passed through a PI controller which generates the reference current for the battery(I_b^*). The actual battery current (I_b) is sensed and is adjusted to the reference value. Hence a cascaded PI controller is required to be tuned. When the battery and the PV source are absent, the dc-link regulation is shifted to the grid control. The switching frequency is selected as 10kHz for the BDC control.

$$I_b^* = (V_{DC}^* - V_{DC}) \times (k_{pd1} + k_{id1}/s) \quad (3)$$

$$e_d = (I_b^* - I_b) \times (k_{pd2} + k_{id2}/s) \quad (4)$$

Here k_{pd1} and k_{id1} are the P and I gains of PI controller 1 and k_{pd2} and k_{id2} are the P and I constants of PI controller 2. e_d is the error signal of second PI.

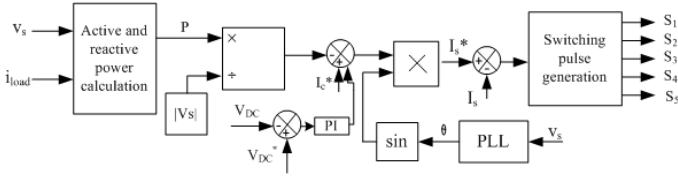


Fig. 4: Control configuration for proposed system

C. MLI Control

This control corresponds to the grid interaction control. Fig. 4 indicates the layout for the grid interactive control. Here, the active component of the load current is estimated and the corresponding power demand is met. A fixed power is fed to the grid. The MLI injects a current (I_c^*) having both active and reactive component so that the reactive power demand of the load can be fulfilled by the MLI to ensure a UPF operation at the grid. Switching logic for the MLI are generated using phase opposition disposition (POD) PWM technique at a switching frequency of 1kHz. Four carrier signals at switching frequency and the reference sinusoidal signal are used for generating the required pulses to the MLI. When the PV and BESS are absent, the dc-link voltage regulation shifts to the inverter control. In this case, the extra power required to maintain the dc-link voltage is also absorbed from the grid.

IV. SIMULATION RESULTS

The simulation study of the system was carried out Matlab/Simulink platform. The PV panel is chosen for a peak power of 4.75kW. Various conditions of the system are analysed. The grid conditions are selected as 230V, 50 Hz system.

A. Steady-State Performance

In this mode of operation, the system performance is analysed under a constant insolation of 1000 W/m². The operating characteristics corresponding to this condition is shown in Fig. 5. The MPPT algorithm is able to trace the peak power (P_{PV}), and the operating voltage (V_{PV}) and current (I_{PV}) corresponding to the MPP is also shown in the Fig. 5. The PV power is adequate to meet the load demand (P_{load}). A fixed power (P_s) of about 2 kW is injected to the grid and the additional power after this is utilized for battery charging purpose. Hence, the battery power (P_b) is obtained as a negative value. A negative battery current (I_b) indicates the charging state of the battery. The dc-link voltage is tuned at the reference value of 400 V. The grid voltage (v_s) and the grid current (i_s) are shown in Fig. 5 and the UPF operation is achieved. MLI current (I_{MLI}) comprises of active and reactive component, and a five level MLI voltage (V_{MLI}) is generated ensuring the self balancing of input capacitors voltages.

B. Dynamic Performance at Irradiation Variation

In this case, a transition in irradiation is applied at time, $t = 2$ sec from 750 W/m² to 1000 W/m² as illustrated in Fig. 6. When the irradiation is increased, the PV output power increases. The PV voltage is maintained almost constant value

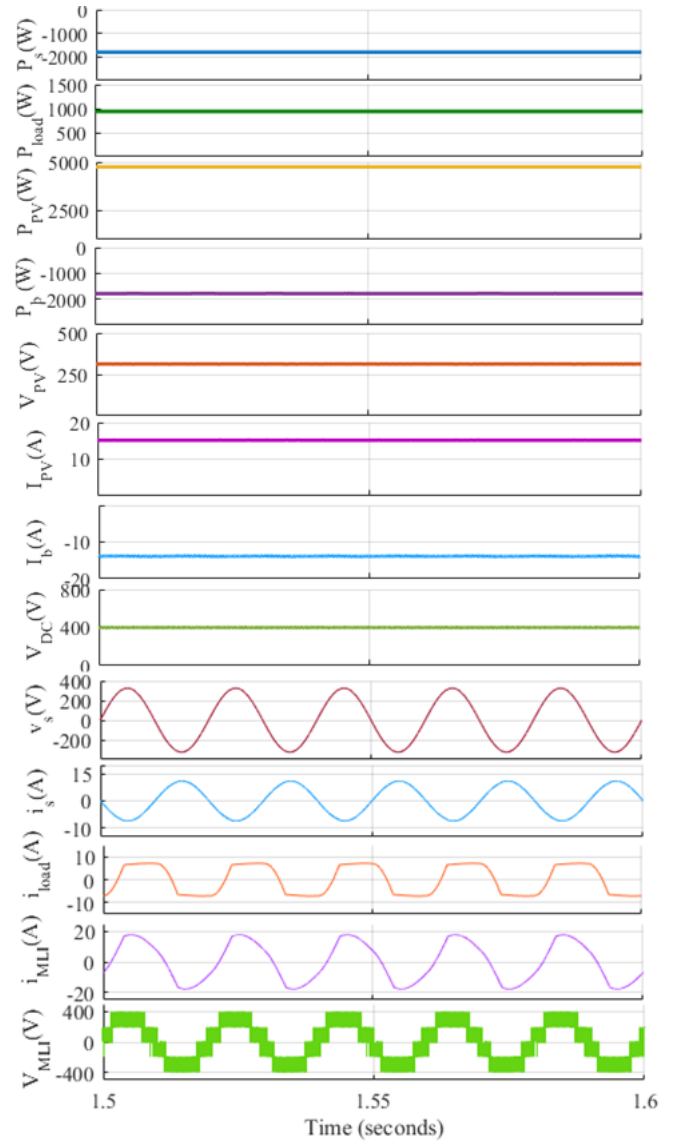


Fig. 5: Steady-state performance of the system at constant irradiance, waveforms of P_s , P_{load} , P_{pv} , P_b , V_{pv} , I_{pv} , I_b , V_{DC} , v_s , i_s , i_{load} , i_{MLI} and V_{MLI}

where as PV current increases. The fixed power is injected to the grid and the load demand is met. With the increase in irradiation, the extra power extracted is used to charge battery hence the charging current increases and so does the battery power. The dc-link voltage is regulated even with the change in irradiation. Since a constant power is injected to the grid, there is no change in the grid power and is maintained at the fixed value. The grid is maintained at UPF operation with grid voltage and current at 180° phase shift indicating that power is injected to the grid.

C. Dynamic Performance at Sudden Disconnection of PV

When the PV is disconnected or when there is no irradiation, the BESS will act as the power source. The performance characteristics when PV is disconnected at time = 2sec is

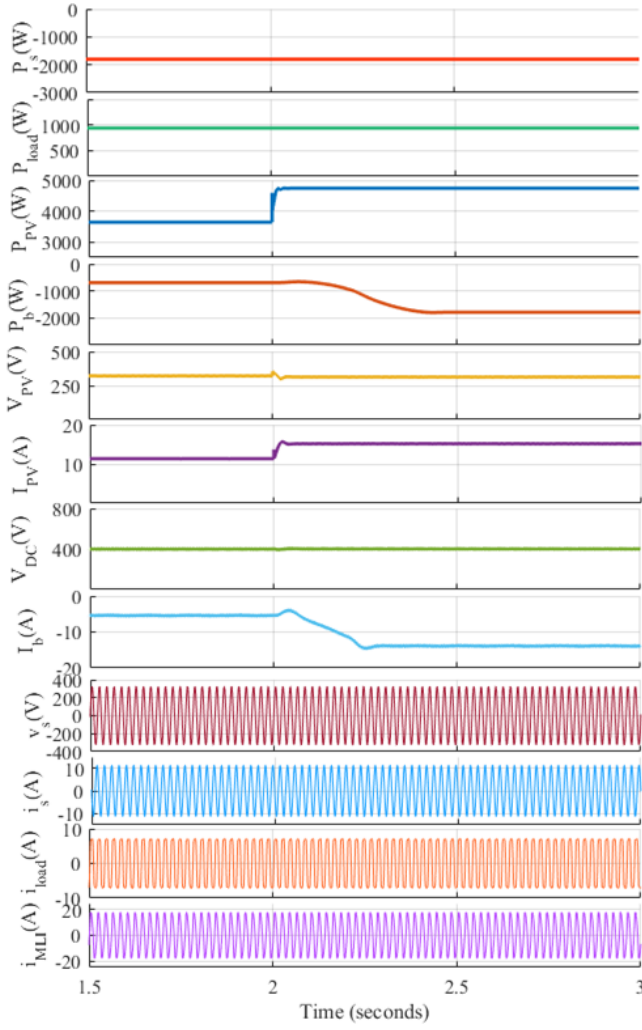


Fig. 6: Dynamic performance of the system at irradiation variation, waveforms of P_s , P_{load} , P_{pv} , P_b , V_{pv} , I_{pv} , I_b , V_{DC} , v_s , i_s , i_{load} , and i_{MLI}

shown in Fig. 7. The irradiation is brought to zero from 1000W/m^2 . The PV current goes to zero and hence the power from the PV is also zero after disconnection. During this scenario, the power required is drawn from the BESS and it is sufficient to attain the load power as well as the fixed power injected to the grid. The regulation of dc-link voltage is also satisfactorily executed even during the dynamic condition. The battery is in the discharging state as indicated by the positive current and power from the battery. The operating waveforms of grid, load and MLI are depicted in Fig. 8. The grid power is maintained at the fixed value and sinusoidal grid waveforms are obtained and the disconnection in the source side does not affect the grid operating condition.

D. DSTATCOM Operation

Fig. 9 demonstrates the performance characteristics of the system under DSTATCOM mode. With the absence of power sources ie, PV and BESS, P_{load} is completely fulfilled the

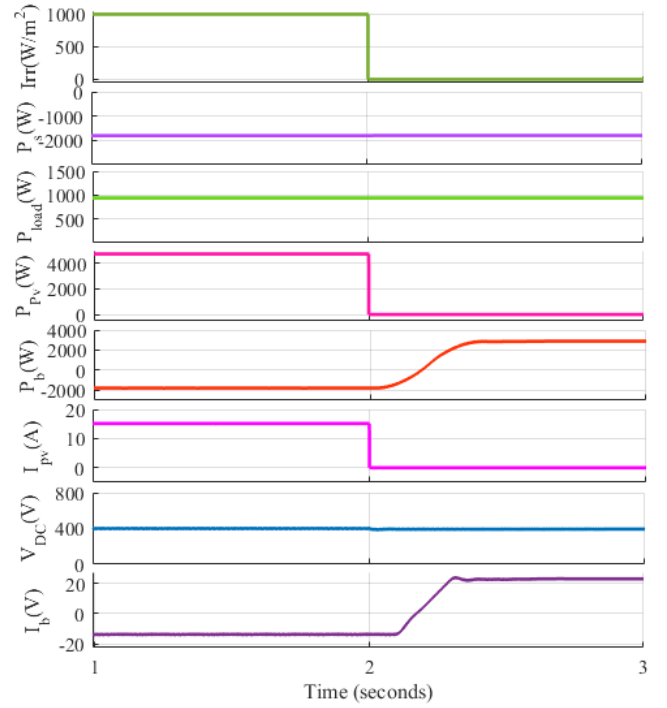


Fig. 7: Dynamic performance of the system at PV disconnection (part-I), waveforms of I_{rr} , P_s , P_{load} , P_{pv} , P_b , I_{pv} , V_{DC} and I_b

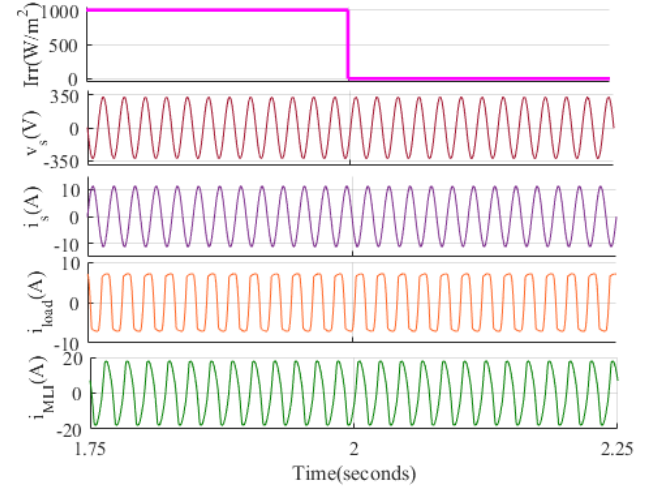


Fig. 8: Dynamic performance of the system at PV disconnection (part-II), waveforms of I_{rr} , v_s , i_s , i_{load} , and i_{MLI}

power from the grid. In this mode, the DSTATCOM operation is performed by the MLI, whereas the reactive power demand of the load (Q_{load}) is met by the MLI (Q_{MLI}), hence the reactive power requirement of the grid (Q_s). v_s and i_s are in phase, so that UPF operation is attained. The control scheme maintains the dc-link voltage regulated. Fig. 10 (a) displays the harmonic spectra of i_{load} . The THD value in the load current is high, 22.95 %. The nonlinear load causes the higher THD. Fig. 10 (b) shows the THD spectra of grid current with a THD

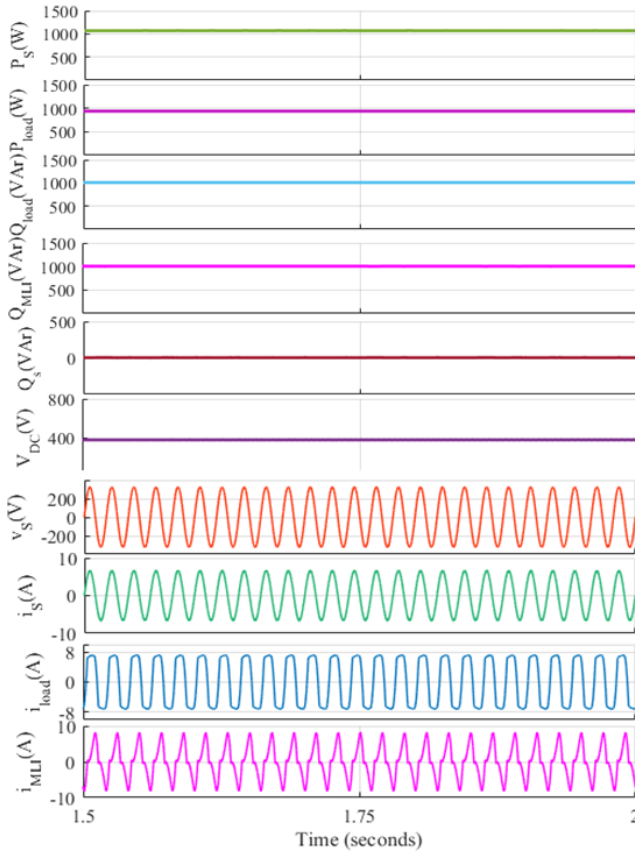


Fig. 9: Waveforms of P_s , P_{load} , Q_{load} , Q_{MLI} , Q_s , V_{DC} , v_s , i_s , i_{load} , and i_{MLI} at DSTATCOM mode

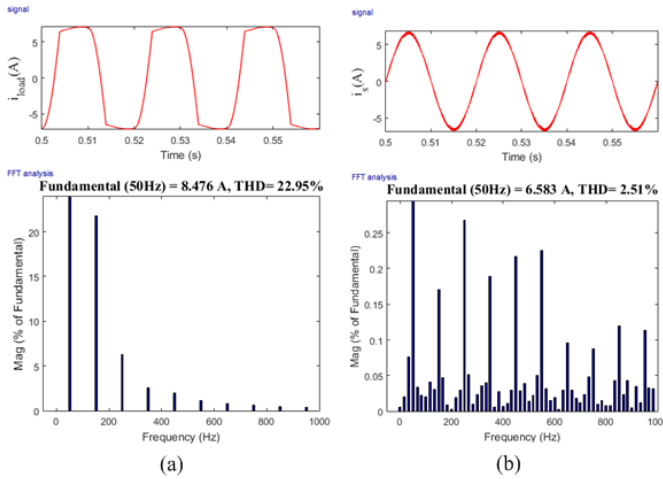


Fig. 10: Harmonic analysis of (a) load current (b) source current

of 2.51 % and is in acceptance with the IEEE-519 standard.

V. CONCLUSION

A five-level inverter-based constant power injection to the grid from the solar PV along with battery support is explored with valid simulation results in this paper. The maximum

power is harvested from the PV panels using the MPPT technique, and it is injected to the grid. The MLI's ability to produce a higher number of levels with lower frequency switching has reduced the output voltage THD and increased the system performance compared with the conventional two-level inverters. Moreover, the charging and discharging of the battery bank is effectively controlled with the cascaded PI controller. The control scheme is found to be satisfactory in all stages. The proposed system also supports a nonlinear local load at the common coupling, and the grid current is maintained as sinusoidal and in-phase with the grid voltage. Hence, the proposed system is suitable for residential as well as commercial applications.

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